COMP-2140: Computer Languages, Grammars, and Translators

Jianguo Lu

School of Computer Science, University of Windsor

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Formal definition of LANGUAGE

- What is a **language**?

Formal definition of LANGUAGE

- What is a **language**? A language is a set of strings
- Examples:
  - English language: $\{$"the brown dog likes a good car", $\ldots\}$
  - Java language: $\{\textit{program} \mid \text{program written in Java}\}$
  - HTML language: $\{\textit{document} \mid \text{document written in HTML}\}$

Formal definition of LANGUAGE

- What is a **language**? A language is a set of strings
- Examples:
  - English language: { "the brown dog likes a good car", ... }
  - Java language: $\{$*program* $|$ program written in Java$\}$
  - HTML language: $\{$*document* $|$ document written in HTML$\}$
- How to define a language?
- It is unlikely that you can enumerate all the sentences, programs, or documents

How to Define a Language

- How to define English

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- A set of words, such as brown, dog, like
- A set of rules
  - A sentence consists of a subject, a verb, and an object;
  - The subject consists of an optional article, followed by an optional adjective, and followed by a noun;
  - ... ...

How to Define a Language

- How to define English
- A set of words, such as brown, dog, like
- A set of rules
  - A sentence consists of a subject, a verb, and an object;
  - The subject consists of an optional article, followed by an optional adjective, and followed by a noun;
  - ... ...
- More formally:
  - Words: {a, the, brown, friendly, good, book, refrigerator, dog, car, sings, eats, likes}
  - Rules:

$$\text{SENTENCE} \rightarrow \text{SUBJECT VERB OBJECT} \tag{1}$$

$$\text{SUBJECT} \rightarrow \text{ARTICLE ADJECTIVE NOUN} \tag{2}$$

$$\text{OBJECT} \rightarrow \text{ARTICLE ADJECTIVE NOUN} \tag{3}$$

$$\text{ARTICLE} \rightarrow a \mid \text{the} \mid \text{EMPTY} \tag{4}$$

$$\text{ADJECTIVE} \rightarrow \text{brown} \mid \text{friendly} \mid \text{good} \mid \text{EMPTY} \tag{5}$$

$$\text{NOUN} \rightarrow \text{book} \mid \text{refrigerator} \mid \text{dog} \mid \text{car} \tag{6}$$

$$\text{VERB} \rightarrow \text{sings} \mid \text{eats} \mid \text{likes} \tag{7}$$

How to decide whether a string is a sentence in the language?

Whether

```
the brown dog likes a good car
```

is a sentence of the language?

Rules

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Derivation of a Sentence

$$\text{SENTENCE} \rightarrow \text{SUBJECT VERB OBJECT} \tag{1}$$
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SENTENCE

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SENTENCE → SUBJECT VERB OBJECT (by rule 1)

Derivation of a Sentence

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Derivation of a Sentence

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&\to \text{the ADJECTIVE NOUN VERB OBJECT} && \text{(by rule 4)}\\
&\to \text{the brown NOUN VERB OBJECT} && \text{(by rule 5)}
\end{aligned}$$

Derivation of a Sentence

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SENTENCE	→ SUBJECT VERB OBJECT	(by rule 1)
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	→the brown dog VERB OBJECT	(by rule 6)

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→the brown dog likes OBJECT	(by rule 7)
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$\to^{+}$the brown dog likes a good car	

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&\rightarrow^{+} \text{the brown dog likes a good car} && \text{(by rules 4,5, 6)}
\end{aligned}
$$

Parse tree of a sentence: The top-down approach

The derivation process can be recorded as a parse tree

`the brown dog likes a good car`

SENTENCE

the brown dog likes a good car

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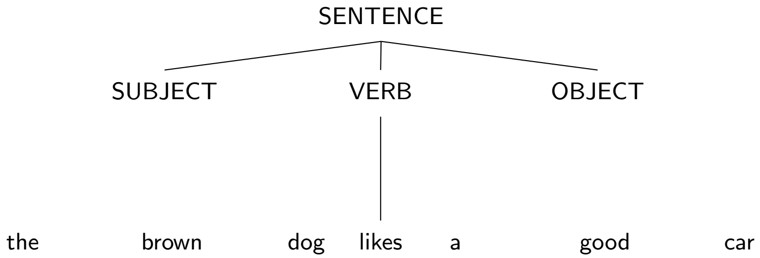

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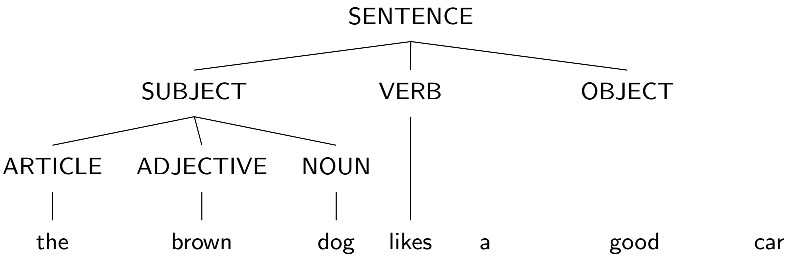

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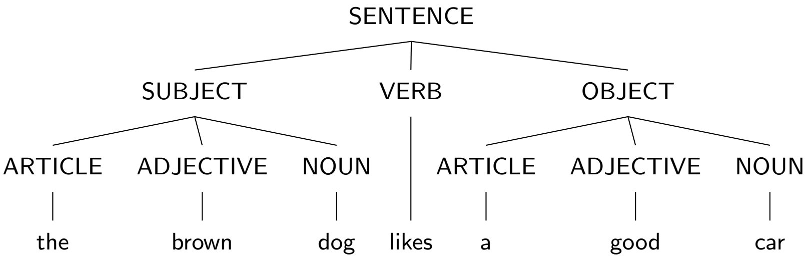


Bottom-up approach

Apply the rules backwards

SENTENCE

the brown dog likes a good car

Bottom-up approach

Apply the rules backwards

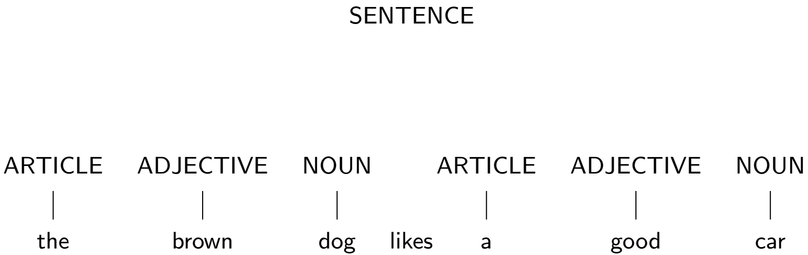
SENTENCE

ARTICLE ADJECTIVE NOUN

the brown dog likes a good car

Bottom-up approach

Apply the rules backwards



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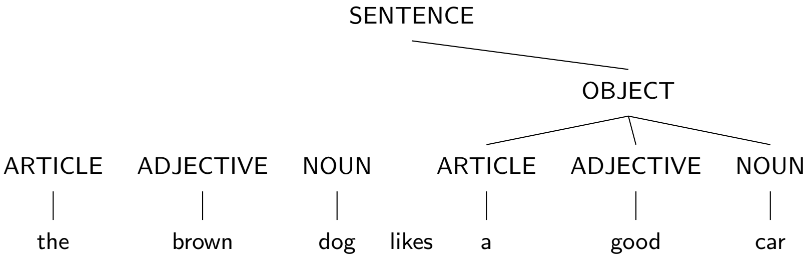

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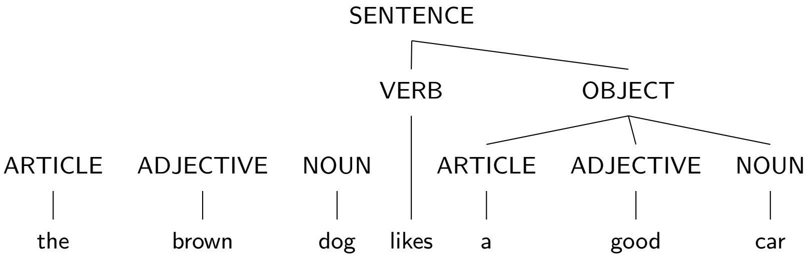

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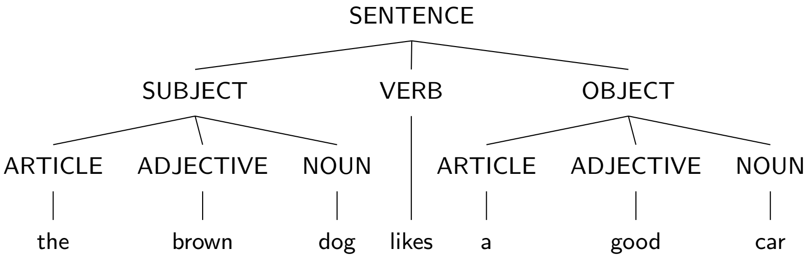

Types of Parsers

- Top-down
  - Repeatedly rewrite the start symbol
  - Find the left-most derivation of the input string
  - Easy to implement
- Bottom-up
  - Start with the tokens and combine them to form interior nodes of the parse tree
  - Find a right-most derivation of the input string
  - Accept when the start symbol is reached
  - Bottom-up is more prevalent

Rules

$$\text{SENTENCE} \rightarrow \text{SUBJECT VERB OBJECT} \tag{1}$$
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- There are two types of symbols

  terminals : e.g., **dog, likes ...**

  non-terminals : e.g., **SUBJECT, VERB, ...**

Formal Definition of Grammar

- A grammar is a 4-tuple $G = (\Sigma, N, P, S)$

$$\begin{aligned} \Sigma &: \text{a finite set of terminal symbols} \\ N &: \text{a finite set of nonterminal symbols} \\ P &: \text{a set of productions} \\ S &: \text{start symbol} \end{aligned}$$

- The English sentence example
  - $\Sigma = \{a, \text{the}, \text{brown}, \text{friendly}, \text{good}, \text{book}, \text{refrigerator}, \text{dog}, \text{car}, \text{sings}, \text{eats}, \text{likes}\}$
  - $N = \{\text{SENTENCE}, \text{SUBJECT}, \text{VERB}, \text{NOUN}, \text{OBJECT}, \text{ADJECTIVE}, \text{ARTICLE}\}$
  - $P = \{\text{rule 1) to rule 7)}\}$
  - $S = \{\text{SENTENCE}\}$

How many sentences can be derived?

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$$\text{SENTENCE} \rightarrow^{+} \text{ARTICLE ADJECTIVE NOUN VERB ARTICLE ADJECTIVE NOUN}$$

How many sentences can be derived?

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$$combinations = 3 * 4 * 4 * 3 * 3 * 4 * 4 = 6912$$

- Note that EMPTY is a terminal
- Grammars have finite set of symbols and rules.
- How to define a language that has infinite number of sentences?

- How can we define an infinite language with a finite set of words and finite set of rules?
- Using recursive rules:
  - SUBJECT/OBJECT can have more than one adjective:

$$\begin{aligned}\text{SUBJECT} &\rightarrow \text{ARTICLE ADJECTIVES NOUN}\\ \text{OBJECT} &\rightarrow \text{ARTICLE ADJECTIVES NOUN}\\ \text{ADJECTIVES} &\rightarrow \text{ADJECTIVE} \mid \text{ADJECTIVES ADJECTIVE}\end{aligned}$$

- Example sentence:

```
the good brown dog likes a good friendly book
```

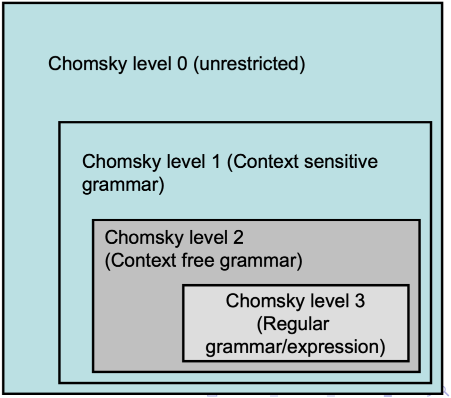
Chomsky hierarchy

- LHS: always have one symbol? Always non-terminals?
- RHS: how many symbols can we have?

$$\text{SENTENCE} \rightarrow \text{SUBJECT VERB OBJECT} \tag{1}$$

(2)

- Level 3: Regular grammar
- Level 2: Context-free grammar
- Level 1: Context-sensitive grammar
- Level 0: Unrestricted grammar



Noam Chomsky Hierarchy

- Noam Chomsky hierarchy is based on the form of production rules.
- **General Form:**

$$\alpha_1\alpha_2\alpha_3\ldots\alpha_n \rightarrow \beta_1\beta_2\beta_3\ldots\beta_m$$

  where α and β are from terminals and non-terminals, or empty.
- **Level 3: Regular Grammar**
  - Form: $\alpha \rightarrow \beta$ or $\alpha \rightarrow \beta_1\beta_2$
  - $n = 1$, and α is a non-terminal.
  - β is either a terminal or a terminal followed by a non-terminal.
  - RHS contains at most one non-terminal at the right end.
- **Level 2: Context-Free Grammar**
  - Form: $\alpha \rightarrow \beta_1\beta_2\beta_3\ldots\beta_m$
  - α is non-terminal.
- **Level 1: Context-Sensitive Grammar**
  - $n < m$. The number of symbols on the LHS must not exceed the number of symbols on the RHS.
- **Level 0: Unrestricted Grammar**

Context-sensitive grammar

- Called context-sensitive because you can construct the grammar of the form
- $A\alpha B \rightarrow A\beta B$
- $A\alpha C \rightarrow A\gamma B$
- The substitution of α depending on the surrounding

Takeaways

- What is language
- Grammars
  - Formal definition, Chomsky hierarchy
  - Regular grammar, lexical analysis
  - Context free grammar, parsing.
- Parser
  - Top down and bottom up
- Language translators
  - Compiler, interpreter

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.
If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.